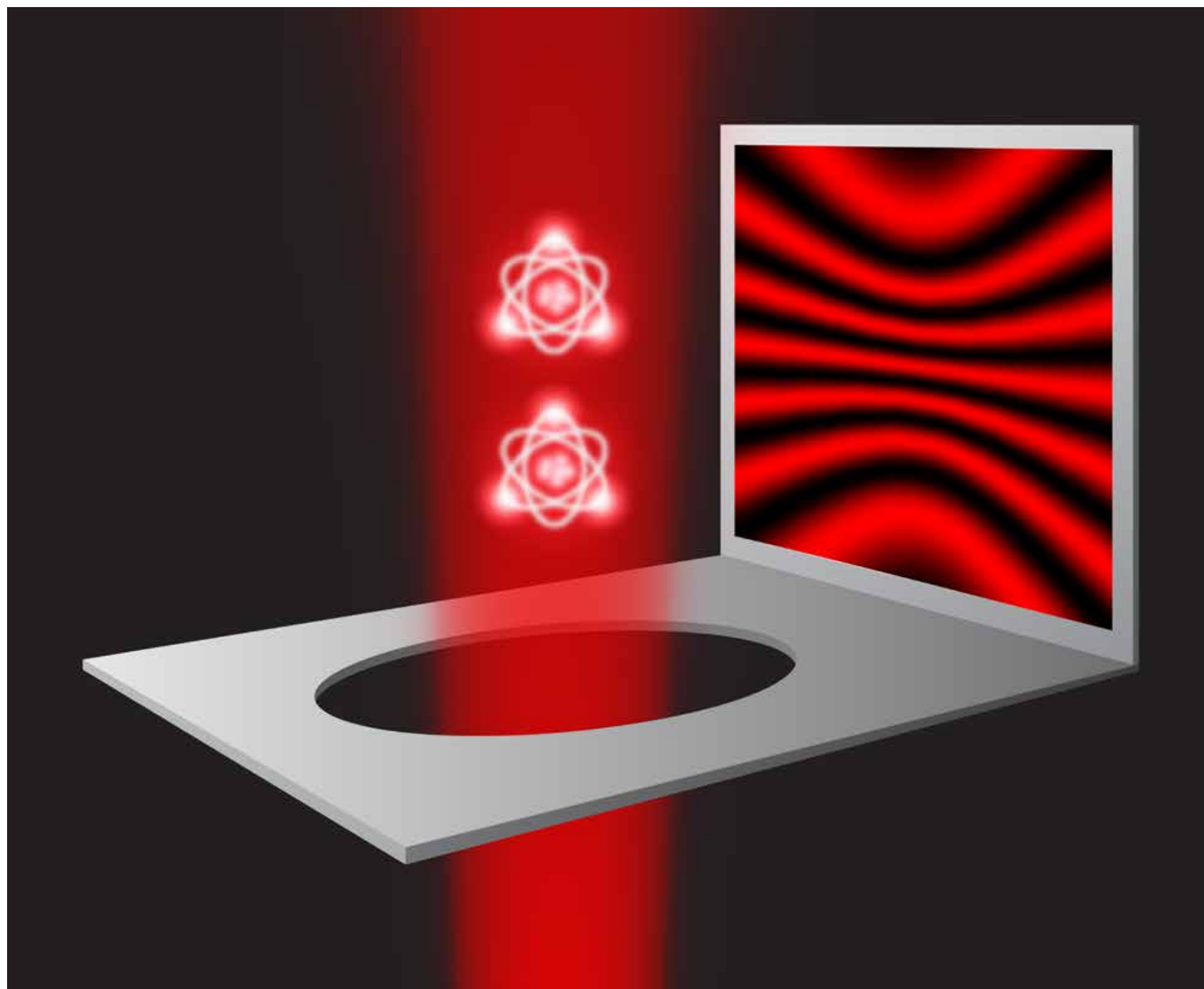
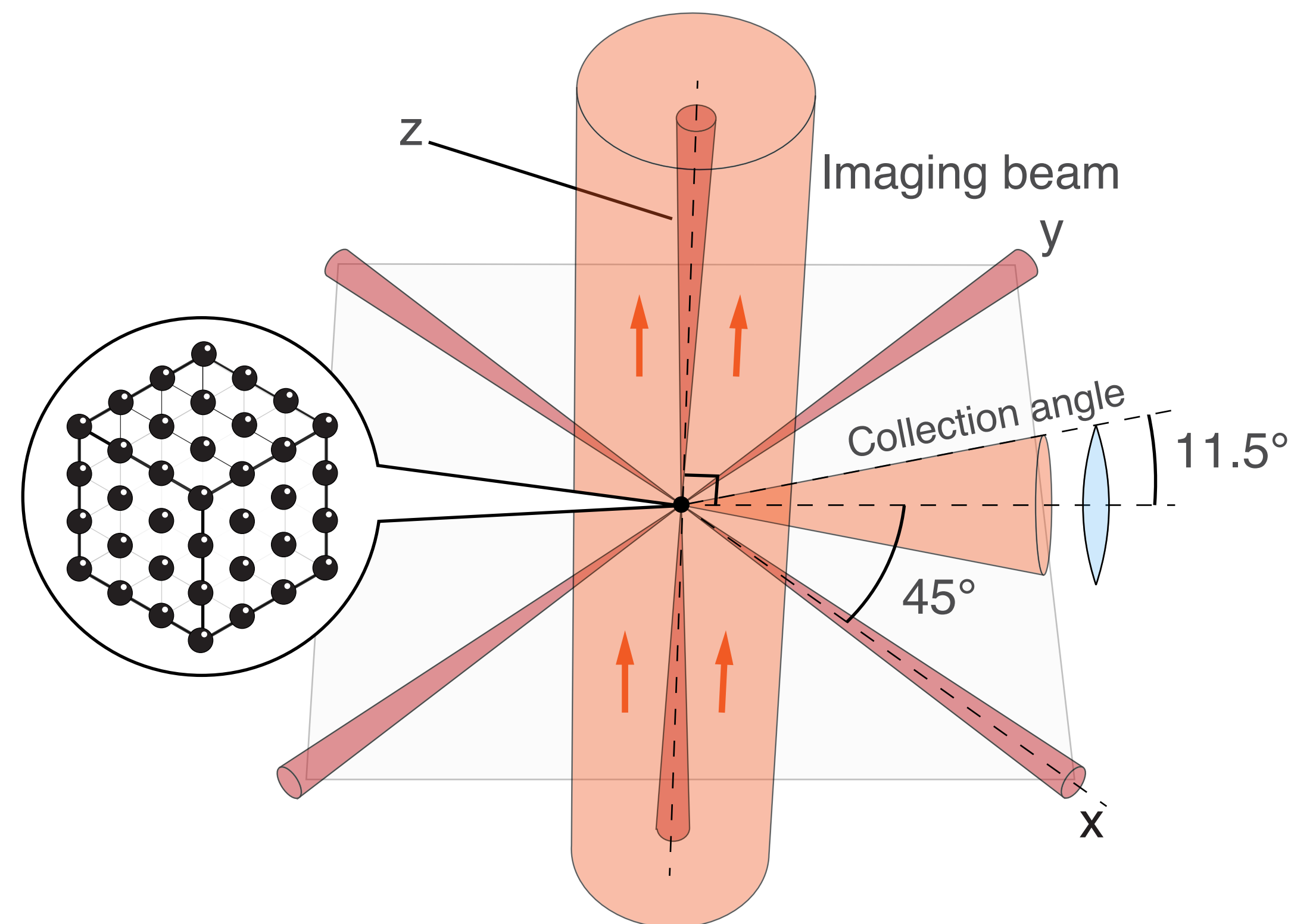


Summary



Physical Review Letters, 135(4), 43601.
Press release on MIT News.



We realize a gedankenexperiment with ultracold atoms in optical lattices to study the fundamental physics of light scattering from single atoms and closely-spaced atom pairs. We see the coherence does not depend on the trapping potential, and observe evidence for long-range dipole-dipole interactions.